

EdgePulse

Edge-AI Predictive Maintenance System

Executive Report

Model	XGBoost
Dataset	NASA IMS Bearing Dataset
Version	v1.0
Generated	21 July 2026, 04:33 PM

AI-powered industrial health monitoring and predictive maintenance.

Executive Summary

Metric	Value
Machines Analysed	100
Model Accuracy	65.0%
Average Confidence	90.2%
Overall Fleet Risk	CRITICAL
Critical Machines	51
Immediate Shutdown	9

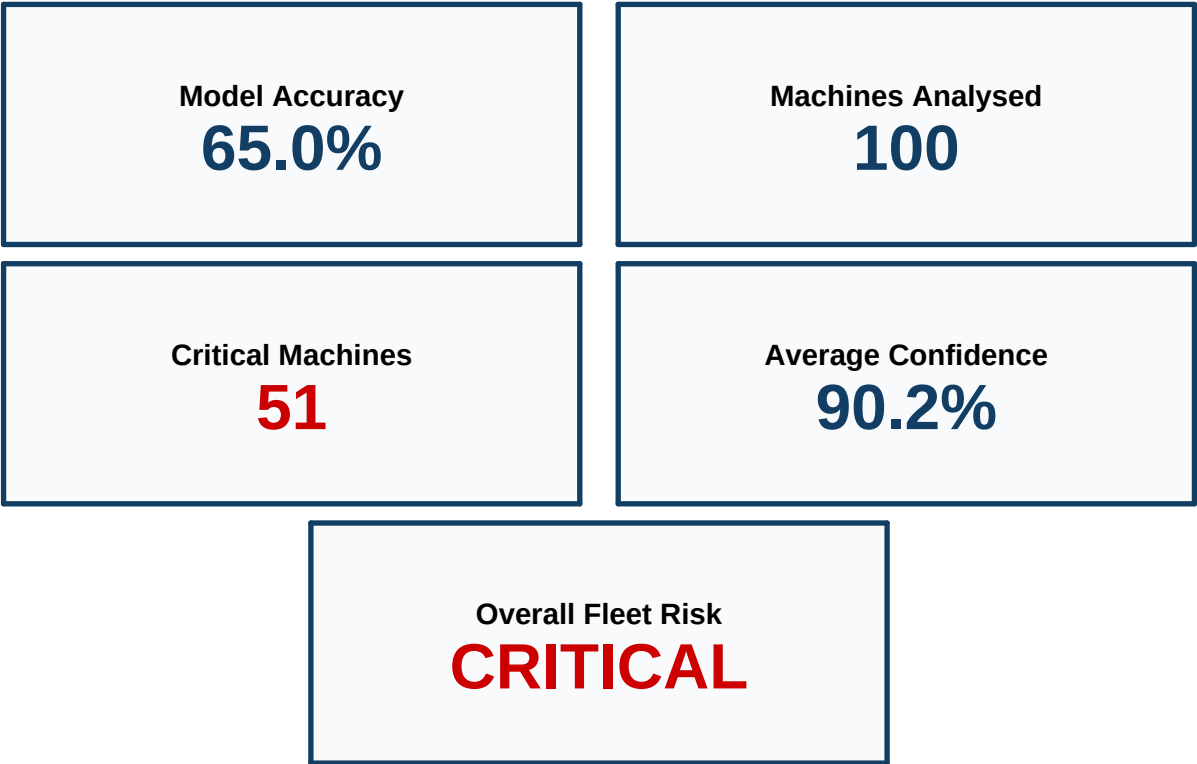
The EdgePulse system analysed **100** rotating machinery samples using an XGBoost-based predictive maintenance model. The model achieved an accuracy of **65.0%** with an average prediction confidence of **90.2%**. Based on the current analysis, the fleet risk level is classified as **CRITICAL**. Preventive maintenance should prioritise critical and imminent failure machines to minimise downtime and operational costs.

- Analysed **100** machine records.

- Detected **51** critical assets.
- **9** machines require immediate shutdown.
- Average prediction confidence: **90.2%**.
- Overall fleet risk: **CRITICAL**.

Model Information

Property	Value
Model	EdgePulse
Algorithm	XGBoost
Dataset	NASA IMS Bearing Dataset
Version	v1.0



Health Distribution

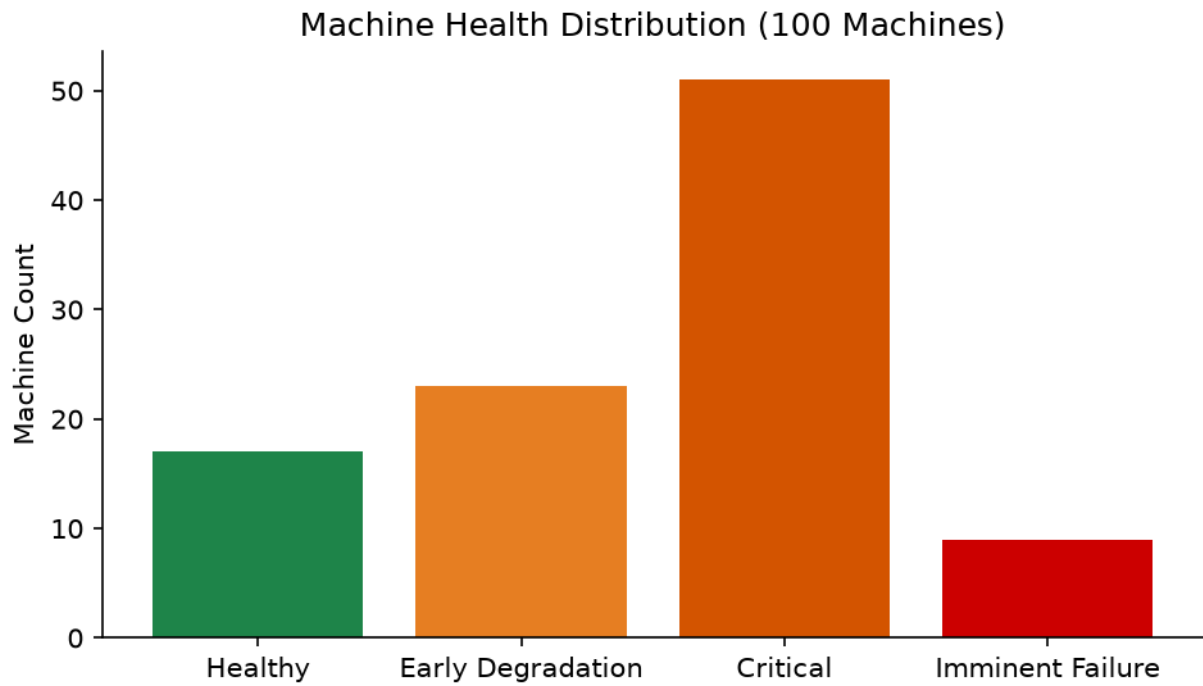


Figure 1. Health Distribution of Predicted Machine Conditions.

Prediction Summary

Health Stage	Machine Count
Healthy	17
Early Degradation	23
Critical	51
Imminent Failure	9

Maintenance Summary

Maintenance Action	Machine Count
Monitor	17
Schedule Inspection	23
Maintain Within 7 Days	51
Immediate Shutdown	9

Maintenance Distribution

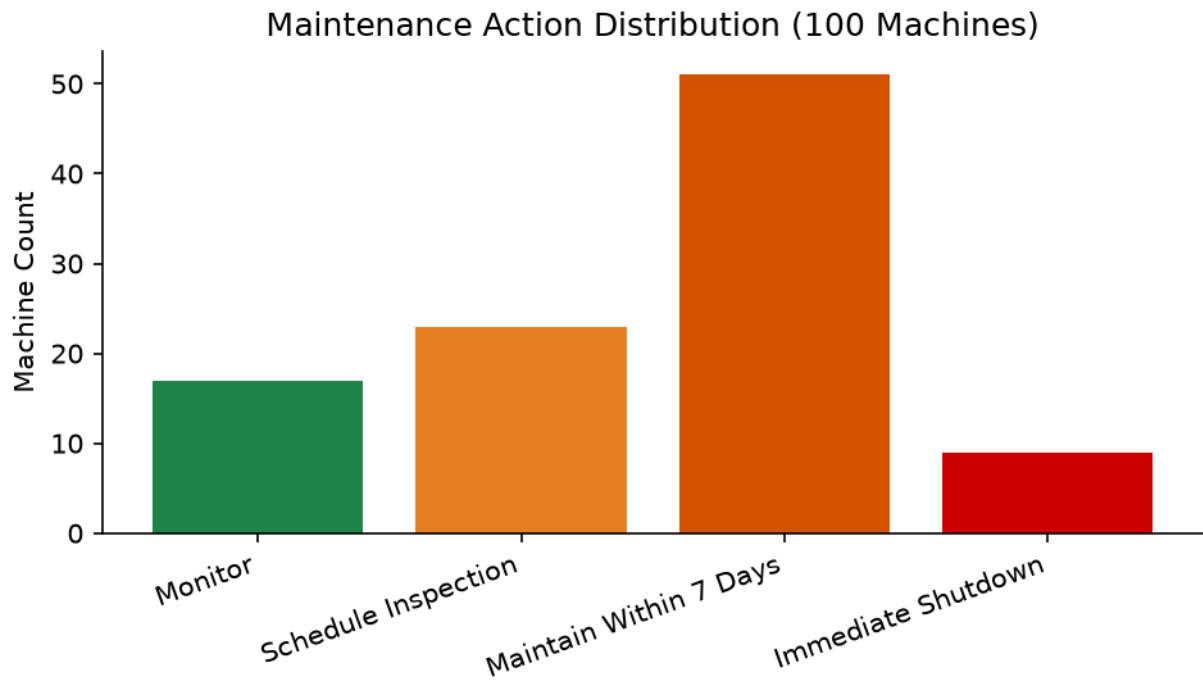


Figure 2. Maintenance Action Distribution Across the Fleet.

Confidence Distribution

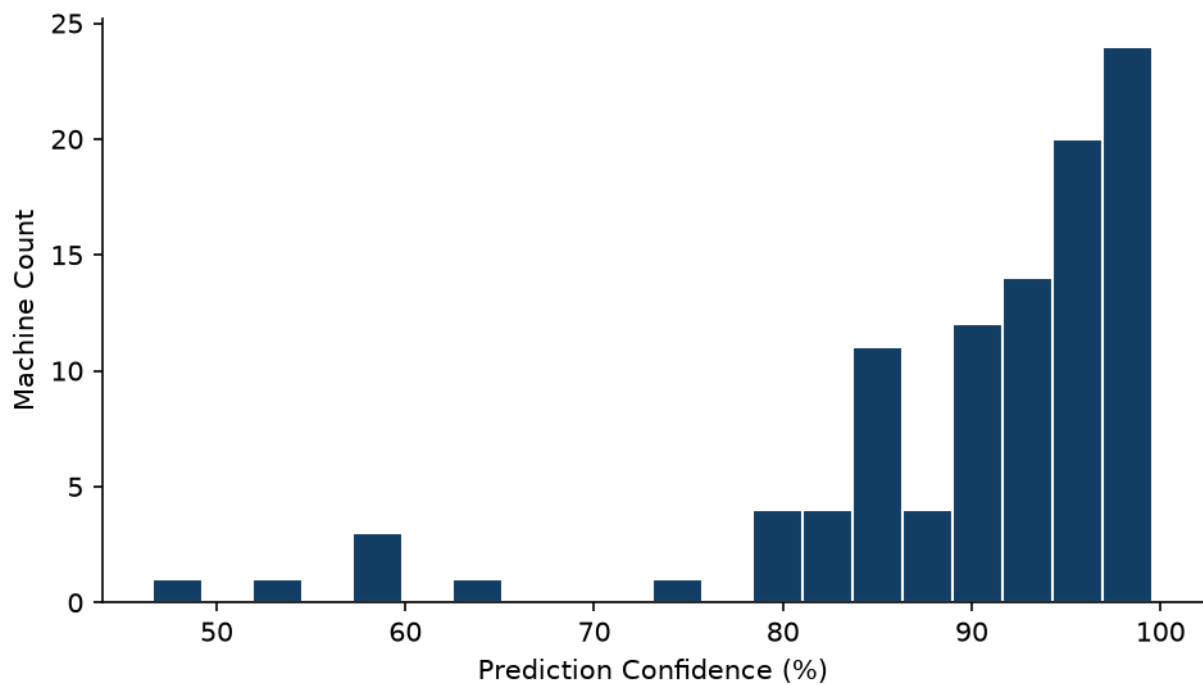


Figure 3. Prediction Confidence Distribution Across Analysed Machines.

Statistic	Value
Average	90.17%
Minimum	46.60%
Maximum	99.61%
Standard Deviation	10.44%
Confidence Quality	HIGH

The model demonstrates high prediction confidence.

Feature Importance

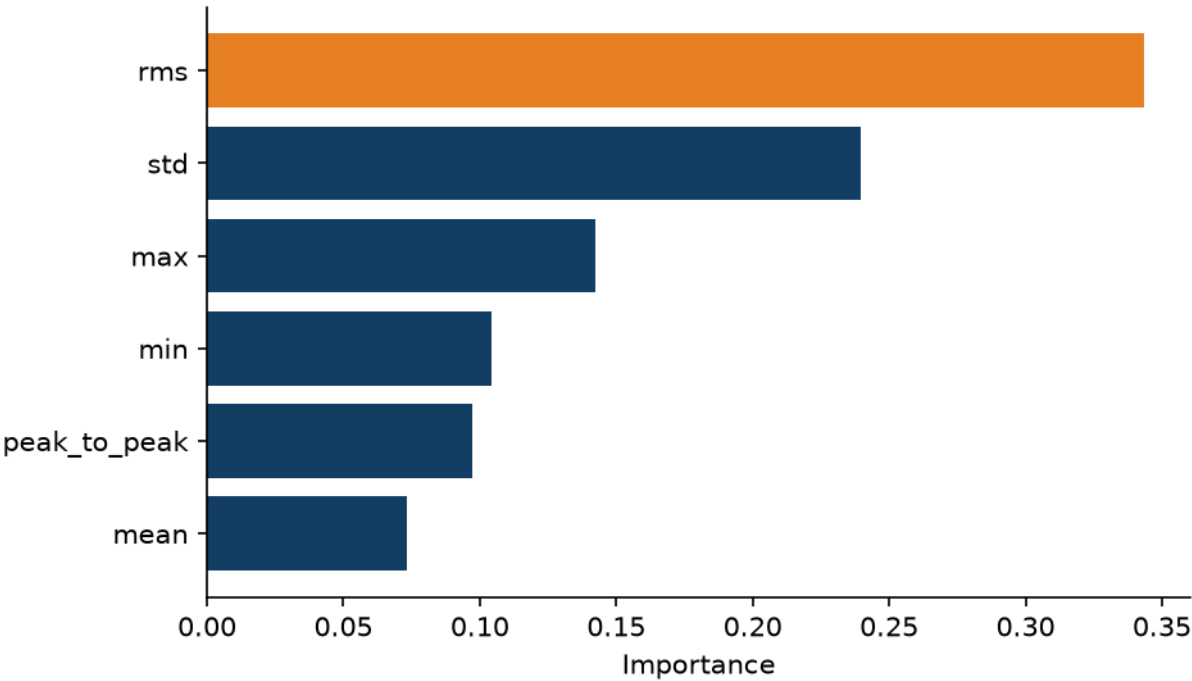


Figure 4. Feature Importance Analysis for the XGBoost Model.

Metric	Value
Top Feature	RMS
Importance	34.3%

High RMS values indicate increased vibration severity and possible bearing degradation.

RMS is the strongest predictor in the XGBoost model, contributing **34.3%** of the total feature importance.

Operational Insights

- Fleet Health: **17.0%** of machines are operating in a healthy state.
- Critical Assets: **51** machines (**51.0%**) require urgent maintenance.
- Immediate Shutdown Required: **9** machines (**9.0%**) are at imminent risk.
- Scheduled Inspection: **23** machines are recommended for preventive inspection.
- **60** machines (**60.0%**) require maintenance within the current planning cycle.
- Model confidence is High (**90.2%**).
- Overall Fleet Risk Level: **CRITICAL**

Immediate shutdown required for high-risk machines.

Maintenance Recommendations

Prediction	Recommended Action	Priority
Healthy	Monitor	Low
Early Degradation	Schedule Inspection	Medium
Critical	Maintain Within 7 Days	High
Imminent Failure	Immediate Shutdown	Critical

Maintenance recommendations are generated automatically from the prediction class mapping, ensuring consistency between model outputs and operational actions.

Conclusion

The EdgePulse predictive maintenance system analysed **100** rotating machinery records using an XGBoost-based machine learning model.

The analysis identified **17** healthy machines (17.0%), **51** critical machines (51.0%), and **9** machines (9.0%) requiring immediate shutdown.

The model achieved an average prediction confidence of **90.2%** and classified the overall fleet risk as **CRITICAL**.

Based on these findings, maintenance efforts should prioritise **60** high-risk machines to minimise unexpected equipment failures and reduce operational downtime.

Business Impact

- Reduced unplanned equipment downtime.
- Improved maintenance scheduling.
- Lower maintenance costs through predictive intervention.
- Enhanced operational reliability and asset utilisation.

Future Scope

- Real-time IoT sensor integration.
- Edge deployment on industrial devices.
- Automated maintenance scheduling.
- Cloud-based fleet monitoring dashboard.

EdgePulse demonstrates how explainable AI and predictive analytics can transform industrial maintenance by enabling proactive decision-making, improving asset reliability, and reducing operational costs.